

SHAMIM AHAMED J

Machine Learning Engineer | AI Engineer | Data Science

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PROFESSIONAL SUMMARY

B.E. (ECE) graduate and Python-focused engineer with hands-on experience across Machine Learning and AI system development. Built a wireless network quality prediction model comparing XGBoost, Decision Tree, and SVM classifiers, covering the full ML workflow from data preprocessing and feature engineering to model evaluation. Developed AI-powered applications using LangChain, LangGraph, and Retrieval-Augmented Generation (RAG) with vector databases (ChromaDB) during a professional Software Engineer experience at Tensaw Technologies, building backend REST APIs with FastAPI to serve AI-driven features. Comfortable working across the Software Development Life Cycle (SDLC), from data handling through backend integration, with strong Python fundamentals and a structured, problem-solving approach. Seeking to apply this Machine Learning, Data Science, and AI engineering foundation to an entry-level ML/AI Engineer role.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript

Machine Learning: XGBoost, Decision Trees, SVM, Data Preprocessing, Feature Engineering, Model Evaluation

AI / LLM Systems: LangChain, LangGraph, OpenAI APIs, Retrieval-Augmented Generation (RAG), ChromaDB (Vector DB), Prompt & Workflow Design

Data Science Tools: Pandas, NumPy (data analysis & preprocessing)

Backend & APIs: FastAPI, REST APIs, Node.js, JWT Authentication

Databases: MySQL, MongoDB

Software Engineering: SDLC, Git-based version control, structured backend architecture, API reliability & testing

Tools: Git, GitHub, VS Code, Postman, Docker

EXPERIENCE

Software Engineer — Tensaw Technologies India Private Limited *May 2025 – May 2026 / Kochi, Kerala*

- Developed features for a clinical AI application using Python, FastAPI, and LangGraph for multi-agent workflow orchestration, following structured SDLC practices.
- Integrated Large Language Models (LLMs) into backend services to support intelligent document processing and clinical decision workflows.
- Designed and consumed REST APIs to connect AI agent pipelines with application data and frontend interfaces.
- Built and tested backend logic in Python, focusing on API reliability, structured data flow between AI agents, and consistent output formatting.
- Gained practical experience in AI application architecture, prompt/workflow design, Git-based version control, and backend software development best practices.

PROJECTS

WirelessIQ — Wireless Network Quality Prediction (Machine Learning Project)

Python · XGBoost · Decision Trees · SVM · Data Preprocessing · Model Evaluation

Objective:

Predict wireless network link quality (Good / Moderate / Poor) from signal parameters using machine learning classification models.

Responsibilities:

- Collected and pre-processed signal data (RSSI, SNR, path loss) and engineered features for model input.
- Trained and compared multiple classification models — XGBoost, Decision Trees, and SVM — to identify the best-performing approach.
- Evaluated model performance using accuracy, precision, recall, and confusion matrix analysis.

Achievements:

- Achieved high classification accuracy for network quality prediction using the tuned XGBoost model, validated against simulated channel conditions.

DocMind — LLM Powered Document Intelligence Agent

LangGraph · LangChain · OpenAI GPT-4o · ChromaDB · FastAPI · Python

Objective:

Build an AI-powered system enabling semantic search and question answering over documents using retrieval-augmented generation.

Responsibilities:

- Designed document retrieval pipelines using the ChromaDB vector database for context-aware responses.
- Implemented multi-agent orchestration using LangGraph and developed FastAPI endpoints to serve AI-powered features.

Achievements:

- Delivered a working end-to-end RAG pipeline capable of accurate, context-grounded question answering over uploaded documents.

ClinicCore — FastAPI Clinic Management System

FastAPI · Async SQLAlchemy · PostgreSQL · MySQL · JWT · Python

Objective:

Build a secure, modular backend API to manage clinics, doctors, patients, and appointments.

Responsibilities:

- Implemented JWT authentication and Role-Based Access Control (RBAC) for secure API access.
- Designed a layered backend architecture using async SQLAlchemy, including appointment scheduling and reusable backend services.

Achievements:

- Delivered a fully functional, secure, multi-role backend API supporting real-world clinic operations.

EDUCATION

Bachelor of Engineering (Electronics and Communication Engineering) — PSNA College of Engineering and Technology 2019 – 2023 | CGPA: 8.15 / 10

HSC — St. Mary's Higher Secondary School 2018 – 2019 | 87.5%

SSLC — Annai Teresa Matric Higher Secondary School 2016 – 2017 | 94.6%

CERTIFICATIONS

- Claude Code in Action — Anthropic
- Full Stack Crash Course — Udemy, Oct 2024 (React.js, HTML, CSS, JavaScript, Bootstrap, MySQL)